

Java Dequeue

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In computer science, a double-ended queue (deque, often abbreviated to deque, pronounced deck) is an abstract data type that generalizes a queue, for which elements can be added to or removed from either the front (head) or back (tail).

Deque interfaces can be implemented using various types of collections such as `LinkedList` or `ArrayDeque` classes. For example, deque can be declared as:

```
Deque deque = new LinkedList<>();
or
Deque deque = new ArrayDeque<>();
```

You can find more details about Deque [here](#).

In this problem, you are given N integers. You need to find the maximum number of unique integers among all the possible contiguous subarrays of size M .

Note: Time limit is 3 second for this problem.

Input Format

The first line of input contains two integers N and M : representing the total number of integers and the size of the subarray, respectively. The next line contains N space separated integers.

Constraints

$$1 \leq N \leq 100000$$

$$1 \leq M \leq 100000$$

$$M \leq N$$

The numbers in the array will range between $[0, 10000000]$.

Author [akashs_csedu](#)

Difficulty **Medium**

Max Score 20

Submitted By 95808

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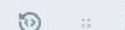
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```
1 import java.util.*;
2
3 public class Solution {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int n = sc.nextInt();
8         int m = sc.nextInt();
9
10        int[] arr = new int[n];
11
12        for (int i = 0; i < n; i++) {
13            arr[i] = sc.nextInt();
14        }
15
16        Map<Integer, Integer> frequency = new HashMap<>();
17
18        int maxUnique = 0;
19
20        // First window
21        for (int i = 0; i < m; i++) {
22            frequency.put(arr[i], frequency.getOrDefault(arr[i], 0) + 1);
23        }
24
25        maxUnique = frequency.size();
26
27        // Slide the window
28        for (int i = m; i < n; i++) {
29
30            // Add new element
31            frequency.put(arr[i], frequency.getOrDefault(arr[i], 0) + 1);
32
33            // Remove element leaving the window
34            int outgoing = arr[i - m];
35
36            frequency.put(outgoing, frequency.get(outgoing) - 1);
37
38            if (frequency.get(outgoing) == 0) {
```

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 Sample Test case 0

Input (stdin)

Download

1	6 3
2	5 3 5 2 3 2

Your Output (stdout)

1	3
---	---

Expected Output

Download

1	3
---	---

Java Hashset

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In computer science, a set is an abstract data type that can store certain values, without any particular order, and no repeated values(Wikipedia). $\{1, 2, 3\}$ is an example of a set, but $\{1, 2, 2\}$ is not a set. Today you will learn how to use sets in java by solving this problem.

You are given n pairs of strings. Two pairs (a, b) and (c, d) are identical if $a = c$ and $b = d$. That also implies (a, b) is not same as (b, a) . After taking each pair as input, you need to print number of unique pairs you currently have.

Complete the code in the editor to solve this problem.

Input Format

In the first line, there will be an integer T denoting number of pairs. Each of the next T lines will contain two strings seperated by a single space.

Constraints:

- $1 \leq T \leq 100000$
- Length of each string is atleast 5 and will consist lower case letters only.

Output Format

Print T lines. In the i_{th} line, print number of unique pairs you have after taking i_{th} pair as input.

Sample Input

```
5
john tom
john mary
john tom
mary anna
mary anna
```

Author

Shafaet

Difficulty

Easy

Max Score

10

Submitted By

131344

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```
1 import java.io.*;
2 import java.util.*;
3 import java.text.*;
4 import java.math.*;
5 import java.util.regex.*;
6
7 public class Solution {
8
9     public static void main(String[] args) {
10         Scanner s = new Scanner(System.in);
11         int t = s.nextInt();
12         String [] pair_left = new String[t];
13         String [] pair_right = new String[t];
14
15         for (int i = 0; i < t; i++) {
16             pair_left[i] = s.next();
17             pair_right[i] = s.next();
18         }
19
20         Set<String> set = new HashSet<>();
21
22         for (int i = 0; i < t; i++) {
23             String pair = pair_left[i] + " " + pair_right[i];
24
25             set.add(pair);
26
27             System.out.println(set.size());
28         }
29
30         s.close();
31     }
```

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Run Code

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✓ Sample Test case 0

Input (stdin)

Download

```
1 5
2 john tom
3 john mary
4 john tom
5 mary anna
6 mary anna
```

Your Output (stdout)

```
1 1
2 2
3 2
```